

Mikser Quarterly Report

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Mikser Core Team

v1

Executive Summary

This document demonstrates Mikser's PDF postprocessor. It is authored in Markdown with YAML front-matter, rendered to HTML by the `report.html-pdf.hbs` layout, and then automatically converted to a print-ready PDF by the `post-pdf` plugin using a headless Chromium browser.

The same source file produces two output artefacts — `report.html` (intermediate) and `report.pdf` (final). The intermediate HTML file is deleted after the PDF is written.

Plugin Pipeline

The pipeline that produces this document involves five plugins:

Step	Plugin	Responsibility
1	<code>front-matter</code>	Parse YAML front-matter from <code>.md</code> files
2	<code>layouts</code>	Match document to layout, set render options
3	<code>render-hbs</code>	Render the Handlebars template to HTML
4	<code>post-pdf</code>	Convert the rendered HTML to PDF via Puppeteer
5	<code>layouts</code> (onComplete)	Write PDF to disk, delete intermediate HTML

How PDF Generation Works

Layout naming convention

The layout filename `report.html-pdf.hbs` encodes two pieces of information:

- `.hbs` — the template engine (Handlebars)

- **.html-pdf** — the output format is HTML with the **pdf** postprocessor

The **getFormatInfo** function in the layouts plugin parses this into `{ format: 'html', postprocessor: 'pdf', template: 'hbs' }`.

Auto-promotion

After a successful render, the **onBeforePostprocess** hook in **engine.js** reads all RENDER journal entries that have **options.postprocessor** set and queues corresponding POSTPROCESS entries. No manual wiring is needed.

Browser lifecycle

The **post-pdf** plugin maintains a single shared Puppeteer browser instance:

```
// setup – called once per postprocess phase
browser = await puppeteer.launch({ headless: true })

// teardown – deferred by 60 s to allow watch-mode reuse
teardownTimer = setTimeout(() => browser.close(), 60_000)
```

If a new build starts within the 60-second window, the timer is cancelled and the existing browser is reused, avoiding a cold launch on every save.

Configuration Reference

Override defaults in **mikser.config.js** under the **post-pdf** key:

```
export default {
  plugins: ['documents', 'layouts', 'render-hbs', 'post-pdf'],
  'post-pdf': {
    launch: { executablePath: '/usr/bin/chromium' },
    navigation: { waitUntil: 'domcontentloaded' },
    pdf: {
      format: 'A4',
      printBackground: true,
      margin: { top: '2cm', bottom: '2cm', left: '2.5cm', right: '2.5cm' }
    },
    teardownDelay: 30_000
  }
}
```

```
}  
}
```

Output Summary

Artefact	Path	Notes
Intermediate HTML	out/en/report.html	Written by render phase; deleted after PDF is produced
Final PDF	out/en/report.pdf	Print-ready A4, background graphics enabled